06/08/24

Advanced Algorithm (PB).

09

NP completeness

10 Algorithm design strategies

Algorithm features

11

finiteness

12 definiteness- each step define one thing

Correctness

01

Algorithm performance - complexity of Algo

02 growth rate of the function by O, θ, Ω

parallel and distributed algorithms

03

Design and analysis of algorithm

04 Greedy algorithm

dynamic programming

05

divide and conquer

Algorithm correctness is proved by induction

in divide an conquer

Recurrence relation of divide and conquer

$$T(n) = D(n) + C(n) + aT(n/b)$$

Binary Search $T(n) = O(1) + T(n/2)$

$$= O(\log_2 n)$$

Merge Sort $T(n) = O(1) + 2T(n/2) + O(n)$

$$= \Theta(n \log n)$$

Quick Sort $T(n) = T(n-1) + 1 + O(n)$

$$= O(n^2) \text{ (Worst case)}$$

Average case computation =

$$E(X) = \frac{1}{n} \sum_{i=1}^{n-1} T(i-1) + T(n-i)$$

Median finding Algorithm.

Long integer Multiplication Algorithm

Ex: prims, cruscal,

09

Greedy Algorithm

10 for optimization problem
optimal substructure

Dynamic Programming

11

overlapping subproblem

12 optimal substructure

bottom up process

01

02

03

SUNDAY 05

04

05